

Results for all tasks

1. SVM-Adult-Age

Val.	Div.	Before			After			Impr. (%)		
		IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.
0.00	9.90				6.82	15.84	80.86	30.11	44.36	-0.12
0.05	9.91				7.13	16.22	80.86	26.91	43.03	-0.11
0.10	9.91				7.34	16.36	80.83	24.79	42.54	-0.15
0.15	9.91				7.46	16.47	80.82	23.63	42.13	-0.16
0.20	9.90				7.61	16.59	80.82	22.09	41.71	-0.16
0.25	9.89				7.69	16.94	80.85	21.23	40.50	-0.13
0.30	9.88				7.98	17.32	80.82	18.29	39.17	-0.16
0.35	9.85				8.15	17.34	80.82	16.54	39.08	-0.16
0.40	9.83				8.22	17.48	80.78	15.75	38.60	-0.21
0.45	9.80				8.34	17.50	80.75	14.59	38.52	-0.25
0.50	9.76	9.76	28.47	80.95	8.53	17.79	80.81	12.61	37.50	-0.18
0.55	9.73				8.78	17.95	80.76	10.05	36.95	-0.23
0.60	9.69				8.75	17.88	80.74	10.41	37.18	-0.26
0.65	9.64				9.04	18.08	80.74	7.42	36.50	-0.26
0.70	9.60				9.13	18.10	80.67	6.52	36.41	-0.35
0.75	9.54				9.31	18.16	80.67	4.67	36.21	-0.35
0.80	9.47				9.43	18.10	80.66	3.38	36.42	-0.36
0.85	9.40				9.48	18.26	80.65	2.87	35.85	-0.37
0.90	9.32				9.81	18.25	80.60	-0.54	35.89	-0.43
0.95	9.25				9.89	18.22	80.61	-1.28	35.99	-0.42
1.00	9.17				9.95	17.91	80.58	-1.97	37.10	-0.46
Fairness (IFr)					PCC / Partial-PCC (p-value):			-1.0 (< 0.01) / -0.98 (< 0.01)		
Fairness (valid-IFr)					PCC / Partial-PCC (p-value):			-0.93 (< 0.01) / -0.98 (< 0.01)		
Accuracy					PCC / Partial-PCC (p-value):			-0.96 (< 0.01) / -0.51 (0.02)		

2. SVM-Credit-age

Val.	Div.	Before			After			Impr. (%)		
		IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.
0.00	13.59				6.66	6.17	75.70	-11.40	-21.73	-0.98
0.05	13.46				4.44	4.27	76.15	25.71	15.68	-0.39
0.10	13.50				4.87	4.67	76.00	18.60	7.88	-0.59
0.15	13.56				6.49	5.70	76.10	-8.49	-12.51	-0.46
0.20	13.57				6.47	6.24	75.95	-8.18	-23.11	-0.65
0.25	13.55				6.18	6.06	76.65	-3.33	-19.65	0.26
0.30	13.54				4.92	4.71	76.65	17.73	6.98	0.26
0.35	13.45				5.96	5.75	76.05	0.38	-13.48	-0.52
0.40	13.42				6.78	5.99	76.30	-13.39	-18.18	-0.20
0.45	13.26				6.87	6.13	76.00	-14.95	-20.98	-0.59
0.50	13.35	5.98	5.07	76.45	7.66	6.99	76.15	-28.17	-38.01	-0.39
0.55	13.30				7.48	6.62	76.50	-25.07	-30.64	0.07
0.60	13.19				8.67	7.77	76.15	-44.93	-53.35	-0.39
0.65	13.13				6.71	5.97	76.55	-12.22	-17.81	0.13
0.70	13.07				6.37	5.63	76.35	-6.50	-11.11	-0.13
0.75	12.97				8.26	7.39	76.60	-38.11	-45.92	0.20
0.80	12.85				7.49	6.50	75.70	-25.25	-28.35	-0.98
0.85	12.79				8.57	7.46	76.65	-43.40	-47.33	0.26
0.90	12.67				9.03	7.67	76.35	-51.04	-51.50	-0.13
0.95	12.51				8.28	7.39	76.50	-38.54	-45.99	0.07
1.00	12.39				9.30	7.66	75.95	-55.54	-51.19	-0.65
Fairness (IFr)					PCC / Partial-PCC (p-value):			-0.81 (< 0.01) / -0.40 (0.08)		
Fairness (valid-IFr)					PCC / Partial-PCC (p-value):			-0.76 (< 0.01) / -0.41 (0.07)		
Accuracy					PCC / Partial-PCC (p-value):			0.29 (0.20) / 0.47 (0.04)		

3. SVM-Credit-Gender

Val.	Div.	Before			After			Impr. (%)		
		IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.
0.00	13.25				7.72	8.21	76.25	22.18	18.88	-1.17
0.05	13.26				7.55	8.24	76.35	23.89	18.54	-1.04
0.10	13.24				7.69	8.08	76.85	22.50	20.14	-0.39
0.15	13.27				6.95	7.60	76.70	29.95	24.84	-0.58
0.20	13.18				6.69	7.51	76.30	32.60	25.76	-1.10
0.25	13.16				6.82	7.00	77.05	31.29	30.76	-0.13
0.30	13.15				6.47	6.82	76.10	34.86	32.54	-1.36
0.35	13.13				7.60	7.99	76.35	23.38	20.96	-1.04
0.40	13.08				6.65	7.15	76.85	32.95	29.30	-0.39
0.45	13.07				7.22	7.60	76.50	27.23	24.86	-0.84
0.50	12.97	9.93	10.11	77.15	5.97	6.39	76.65	39.81	36.84	-0.65
0.55	12.98				6.86	7.39	76.20	30.92	26.89	-1.23
0.60	12.90				7.00	7.44	76.95	28.52	26.46	-0.26
0.65	12.72				6.75	7.14	76.10	31.97	29.36	-1.36
0.70	12.72				7.26	7.75	76.85	26.89	23.41	-0.39
0.75	12.55				6.39	6.77	76.95	35.62	33.04	-0.26
0.80	12.49				6.70	6.98	77.10	32.49	31.00	-0.06
0.85	12.34				6.79	6.88	76.65	31.64	32.00	-0.65
0.90	12.22				5.98	6.51	77.00	39.79	35.60	-0.19
0.95	12.10				6.76	6.95	77.20	31.90	31.24	-0.06
1.00	12.00				7.15	7.52	76.85	27.95	25.63	-0.39
Fairness (IFr)					PCC / Partial-PCC (p-value):			0.47 (0.03) / 0.36 (0.11)		
Fairness (valid-IFr)					PCC / Partial-PCC (p-value):			0.58 (< 0.01) / 0.42 (0.07)		
Accuracy					PCC / Partial-PCC (p-value):			0.49 (0.03) / -0.01 (0.97)		

4. MLPC-Adult-Age

Val.	Div.	Before			After			Impr. (%)		
		IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.
0.00	9.42				3.29	10.12	83.15	56.46	25.49	-0.25
0.05	9.46				3.93	10.10	83.63	47.89	25.59	0.32
0.10	9.47				2.90	8.41	82.49	61.53	38.03	-1.04
0.15	9.46				2.66	6.32	81.29	64.69	53.43	-2.48
0.20	9.47				5.48	9.93	83.60	27.36	26.87	0.29
0.25	9.46				5.21	9.86	83.79	30.92	27.38	0.52
0.30	9.45				4.54	9.61	82.45	39.82	29.23	-1.09
0.35	9.44				3.15	7.96	82.96	58.26	41.38	-0.48
0.40	9.41				4.10	9.03	83.83	45.63	33.49	0.57
0.45	9.38				3.02	6.54	80.67	59.94	51.83	-3.23
0.50	9.31	7.55	13.58	83.36	3.28	6.52	81.56	56.50	52.00	-2.15
0.55	9.29				4.47	9.51	83.78	40.74	29.93	0.50
0.60	9.23				3.54	6.91	82.18	53.11	49.09	-1.42
0.65	9.17				3.60	7.70	82.81	52.26	43.31	-0.66
0.70	9.09				4.46	6.38	81.62	40.88	52.98	-2.09
0.75	9.03				5.12	8.59	83.08	32.15	36.73	-0.33
0.80	8.95				4.86	8.76	83.78	35.57	35.49	0.51
0.85	8.86				4.71	8.34	83.19	37.61	38.58	-0.20
0.90	8.76				4.46	9.05	83.13	40.94	33.34	-0.27
0.95	8.66				5.33	9.42	82.81	29.40	30.58	-0.66
1.00	8.55				2.53	3.87	79.65	66.48	71.49	-4.44
Fairness (IFr)					PCC / Partial-PCC (p-value):			-0.25 (0.27) / -0.20 (0.41)		
Fairness (valid-IFr)					PCC / Partial-PCC (p-value):			0.38 (0.09) / 0.22 (0.35)		
Accuracy					PCC / Partial-PCC (p-value):			-0.22 (0.35) / 0.02 (0.94)		

5. MLPC-Adult-Race

Val.	Div.	Before			After			Impr. (%)		
		IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.
0.00	10.14				1.00	2.75	81.52	64.11	51.93	-2.80
0.05	10.13				1.14	3.65	82.73	59.42	36.29	-1.35
0.10	10.13				1.27	3.99	82.65	54.79	30.35	-1.45
0.15	10.12				1.26	4.16	82.64	54.95	27.36	-1.47
0.20	10.10				1.24	3.48	82.43	55.66	39.28	-1.72
0.25	10.08				0.96	2.77	81.61	65.57	51.65	-2.69
0.30	10.05				1.66	3.78	83.04	40.63	34.02	-0.99
0.35	10.03				1.14	3.40	83.12	59.45	40.74	-0.90
0.40	9.97				1.71	3.93	82.92	38.95	31.39	-1.13
0.45	9.94				1.27	3.42	76.73	54.73	40.32	-8.51
0.50	9.88	2.80	5.73	83.87	1.01	3.20	81.46	64.03	44.10	-2.88
0.55	9.83				1.07	3.05	82.12	61.86	46.79	-2.09
0.60	9.78				1.05	2.55	80.57	62.66	55.42	-3.93
0.65	9.72				1.01	2.71	81.39	63.96	52.64	-2.96
0.70	9.68				1.53	3.45	85.53	59.81	39.81	-1.24
0.75	9.58				1.25	3.58	83.67	55.33	37.60	-2.04
0.80	9.49				1.22	2.97	82.47	56.36	48.21	-1.66
0.85	9.42				1.10	2.93	82.29	60.80	48.90	-1.88
0.90	9.33			1.16	3.13	82.99	58.74	45.39	-1.05	
0.95	9.23			1.14	2.92	82.17	59.28	49.11	-2.02	
1.00	9.14			1.22	2.55	81.22	56.41	55.46	-3.15	
Fairness (IFr)		PCC / Partial-PCC (p-value):			0.04 (0.87)			-0.17 (0.46)		
Fairness (valid-IFr)		PCC / Partial-PCC (p-value):			0.50 (0.02)			-0.02 (0.94)		
Accuracy		PCC / Partial-PCC (p-value):			-0.01 (0.95)			-0.17 (0.46)		

9. MLPC-Credit-Gender

Val.	Div.	Before			IFr	After			Impr. (%)		
		IFr	valid-IFr	Acc.		IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.
0.00	12.75					0.82	0.85	59.80	58.67	59.52	-18.08
0.05	12.77					0.77	0.69	50.20	61.56	67.07	-31.23
0.10	12.82					1.22	1.01	62.95	38.57	51.65	-13.77
0.15	12.86					0.34	0.39	66.10	82.75	81.14	-9.45
0.20	12.91					0.67	0.80	71.00	66.27	61.61	-2.74
0.25	12.83					0.48	0.39	58.85	75.89	81.39	-19.38
0.30	12.78					0.29	0.32	62.10	85.45	84.77	-14.93
0.35	12.69					0.46	0.50	61.10	76.77	76.24	-16.30
0.40	12.79					0.65	0.73	64.80	67.32	65.17	-11.23
0.45	12.65					0.08	0.12	49.75	95.81	94.15	-31.85
0.50	12.58	1.99	2.09	73.00		1.17	1.38	52.20	41.01	34.06	-28.49
0.55	12.58					0.77	0.78	64.85	61.53	62.87	-11.16
0.60	12.52					0.72	0.60	64.50	63.63	71.50	-11.64
0.65	12.38					2.00	2.04	68.55	-0.54	2.32	-6.10
0.70	12.39					0.46	0.49	50.80	77.07	76.41	-30.41
0.75	12.20					1.02	1.14	53.10	48.81	45.54	-27.26
0.80	12.11					0.49	0.51	57.75	75.27	75.43	-20.89
0.85	12.02					0.72	0.61	62.55	63.81	70.60	-14.32
0.90	11.88					0.46	0.39	70.35	77.02	81.51	-3.63
0.95	11.78					0.47	0.61	63.30	76.27	71.02	-13.29
1.00	11.64					0.61	0.67	70.55	69.52	68.11	-3.36
		Fairness (IFr)	PCC /	Partial-PCC (p-value):		0.01 (0.96) /	0.01 (0.97)				
		Fairness (valid-IFr)	PCC /	Partial-PCC (p-value):		-0.03 (0.88) /	-0.05 (0.84)				
		Accuracy	PCC /	Partial-PCC (p-value):		0.17 (0.46) /	-0.04 (0.87)				

10. RF-Bank-Age

Val.	Div.	Before			IFr	After			Impr. (%)		
		IFr	valid-IFr	Acc.		IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.
0.00	10.55					6.43	7.10	90.29	76.40	65.27	-0.13
0.05	10.57					6.05	6.44	90.35	77.80	68.49	-0.06
0.10	10.57					6.15	6.05	90.41	77.41	70.43	-0.00
0.15	10.57					6.66	6.63	90.42	75.56	67.59	0.02
0.20	10.56					6.62	5.99	90.39	75.72	70.73	-0.02
0.25	10.56					6.58	5.62	90.47	75.86	72.53	0.08
0.30	10.54					5.80	5.51	90.44	78.70	73.08	0.04
0.35	10.52					6.69	5.64	90.44	75.47	72.43	0.03
0.40	10.49					6.78	5.42	90.51	75.12	73.51	0.11
0.45	10.47					6.40	5.39	90.43	76.53	73.64	0.03
0.50	10.43	27.25	20.45	90.41		6.23	5.18	90.36	77.14	74.68	-0.05
0.55	10.39					6.06	5.08	90.38	77.74	75.18	-0.03
0.60	10.35					6.97	5.27	90.36	74.44	74.21	-0.05
0.65	10.30					6.12	4.93	90.35	77.52	75.90	-0.06
0.70	10.24					6.42	4.93	90.34	76.43	75.90	-0.07
0.75	10.19					7.04	5.16	90.40	74.15	74.79	-0.00
0.80	10.14					6.40	4.87	90.47	76.50	76.20	0.07
0.85	10.07					7.02	4.88	90.41	74.26	76.16	0.00
0.90	9.99					6.98	4.89	90.39	74.38	76.09	-0.02
0.95	9.92					6.69	4.89	90.45	75.44	76.09	0.05
1.00	9.84					6.33	4.90	90.41	76.76	76.07	0.00
		Fairness (IFr)	PCC /	Partial-PCC (p-value):		-0.36 (0.11) /	-0.11 (0.64)				
		Fairness (valid-IFr)	PCC /	Partial-PCC (p-value):		0.90 (< 0.01) /	0.92 (< 0.01)				
		Accuracy	PCC /	Partial-PCC (p-value):		0.14 (0.56) /	0.27 (0.24)				

11. RF-Credit-Age

Val.	Div.	Before			IFr	After			Impr. (%)		
		IFr	valid-IFr	Acc.		IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.
0.00	13.57					17.78	15.01	76.50	20.12	4.84	0.53
0.05	13.59					17.17	15.86	76.55	22.85	-0.53	0.59
0.10	13.61					16.47	16.08	75.75	26.01	-1.95	-0.46
0.15	13.59					17.22	16.07	76.05	22.64	-1.86	-0.07
0.20	13.57					17.31	16.16	76.55	22.20	-2.46	0.59
0.25	13.56					15.37	15.23	76.40	30.93	3.46	0.39
0.30	13.56					18.13	15.89	76.45	18.53	-0.73	0.46
0.35	13.48					18.45	15.62	75.85	17.10	0.96	-0.33
0.40	13.42					17.04	15.96	76.90	23.41	-1.19	1.05
0.45	13.40					18.22	15.97	75.85	18.15	-1.25	-0.33
0.50	13.19	22.25	15.77	76.10		18.98	15.93	76.35	14.70	-0.96	0.33
0.55	13.20					18.59	17.18	76.55	16.48	-8.91	0.59
0.60	13.16					16.67	15.74	76.15	25.08	0.23	0.07
0.65	13.05					18.20	16.04	75.60	18.21	-1.65	-0.66
0.70	12.95					16.81	14.73	76.45	24.46	6.62	0.46
0.75	12.80					17.95	15.48	75.60	19.36	1.89	-0.66
0.80	12.68					17.91	15.02	76.25	19.52	4.80	0.20
0.85	12.56					18.80	16.32	76.55	15.53	-3.47	0.59
0.90	12.39					17.61	15.61	76.45	20.88	1.06	0.46
0.95	12.20					21.48	15.90	76.45	3.50	-0.81	0.46
1.00	12.06					17.95	16.02	76.45	19.33	-1.57	0.46
		Fairness (IFr)	PCC /	Partial-PCC (p-value):		-0.47 (0.03) /	0.02 (0.94)				
		Fairness (valid-IFr)	PCC /	Partial-PCC (p-value):		-0.00 (0.99) /	-0.11 (0.65)				
		Accuracy	PCC /	Partial-PCC (p-value):		0.01 (0.97) /	-0.23 (0.32)				

12. RF-Credit-Gender

Val.	Div.	Before			IFr	After			Impr. (%)		
		IFr	valid-IFr	Acc.		IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.
0.00	13.15					4.57	5.15	76.65	50.07	28.74	-2.48
0.05	13.17					4.34	4.92	76.80	52.60	31.90	-2.29
0.10	13.09					4.52	5.05	76.60	50.55	30.07	-2.54
0.15	13.08					4.66	5.02	76.55	49.02	30.41	-2.61
0.20	13.09					5.08	5.57	76.90	44.41	22.88	-2.16
0.25	13.09					4.58	5.10	77.30	49.92	29.40	-1.65
0.30	12.98					5.13	5.22	76.95	43.92	27.76	-2.10
0.35	13.02					4.42	4.90	76.65	51.65	32.09	-2.48
0.40	13.08					4.78	4.99	76.05	47.71	30.86	-3.24
0.45	12.99					5.10	5.24	77.55	44.23	27.37	-1.34
0.50	12.90	9.15	7.22	78.60		5.03	5.13	76.55	45.00	28.90	-2.61
0.55	12.81					4.96	5.23	76.55	45.81	27.62	-2.61
0.60	12.79					5.35	5.15	75.70	41.49	28.64	-3.69
0.65	12.66					5.32	4.88	76.50	41.86	32.47	-2.67
0.70	12.59					5.86	5.07	76.40	35.97	29.76	-2.80
0.75	12.50					5.97	5.51	77.05	34.73	23.69	-1.97
0.80	12.30					5.70	5.10	76.60	37.65	29.31	-2.54
0.85	12.27					5.71	5.07	76.05	37.57	29.75	-3.24
0.90	12.17					6.87	5.66	76.80	24.88	21.66	-2.29
0.95	11.97					5.86	4.89	77.45	35.97	32.28	-1.46
1.00	11.88					6.17	5.18	76.40	32.56	28.21	-2.80
		Fairness (IFr)	PCC /	Partial-PCC (p-value):		-0.88 (< 0.01) /	-0.34 (0.14)				
		Fairness (valid-IFr)	PCC /	Partial-PCC (p-value):		-0.16 (0.49) /	-0.08 (0.74)				
		Accuracy	PCC /	Partial-PCC (p-value):		-0.11 (0.64) /	-0.22 (0.36)				